Introduction to Nonlinear Dynamics

Course Placement: *Introduction to Nonlinear Dynamics* (SC401) is a science elective course.

Course Format: 3 hours of lectures every week for the 3-credit course.

Course Prerequisites: Two core courses of the Computational Science (Minor/Honours) Programme- *Introductory Computational Physics* (CS201) & *Modelling and Simulation* (CS302).

Course Content: This course introduces nonlinear dynamics to undergraduate engineering students. It first establishes the necessary theoretical and mathematical foundations and then focuses on developing practical problem-solving skills in a wide variety of domains (many of them of engineering, socio-economic, industrial, biological, and environmental relevance). The two broad classes of mathematical problems in this course pertain to one-dimensional and two-dimensional systems. Special emphasis will be laid on the first-order problems, as it provides technically accessible concepts that are nonetheless effective for analyzing a wide range of problems in depth. All the theoretical concepts introduced in one-dimensional systems will be supplemented strongly by practical examples and exercises. Two-dimensional systems are of an advanced nature, but they can be understood easily once the theoretical foundations of one-dimensional systems have been laid.

Text Books:
1.) *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry and Engineering*, Steven H. Strogatz, Levant Books
2.) *Differential Equations and Their Applications: An Introduction to Applied Mathematics*, Martin Braun, Springer-Verlag

Reference Books:
1.) *Nonlinear Ordinary Differential Equations: An Introduction to Dynamical Systems*, D. W. Jordan & P. Smith, Oxford University Press

Assessment Method: The total grade will be based on assignments (30%) and scheduled examinations (70%).

Course Outcomes:
The course is expected to inspire confidence among engineering undergraduates to enable them to solve many practical problems using a basic set of analytical skills within their professional range. After successful completion of the course, the student will have the ability to:

- Analyze problems using first principles of mathematics and physics.
- Synthesize and interpret observational facts of the real world to provide valid conclusions.

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X	X										

LECTURE SCHEDULE

Sl. No.	DESCRIPTION		No. of Lectures
1	**Review Topics**		**5**
	1.1	Methodology of nonlinear dynamics and its broad perspectives.	
	1.2	Differential equations and their dynamic features.	
2	**One-Dimensional Systems**		**20**
	2.1	Autonomous systems, flows on the line and their dynamics.	
	2.2	Fixed points and linear stability analysis.	
	2.3	The logistic equation, tumour growth, Newton's law of cooling, etc.	
	2.4	Bifurcations – saddle-node, transcritical and pitchfork.	
	2.5	Flows on the circle.	
3	**Two-Dimensional Systems**		**15**
	3.1	Coupled autonomous systems. Phase portraits.	
	3.2	Linearization, fixed points and their classification.	
	3.3	Systems in social and biological dynamics: love affairs, arms race, predator-prey interactions, competitive exclusion, etc.	
	3.4	Conservative and reversible systems.	
	3.5	Limit cycles. Bifurcation in two-dimensional systems.	
	3.6	Introduction to chaotic systems.	